13 LCM & HCF OF ALGEBRAIC FACTORS

PRACTICE SHEET

1. What is the HCF of polynomials $x^3+3x^2y+2xy^2$ and $x^4+6x^3y+8x^2y^2$?
(1) x(x+2y) (2)x(x+3y)
(3) (x+2y (4) None of these

2. If the HCF of x^3+mx^2-x+2m and x^2+mx-2 is a linear polynomial, then what is the value of m?
(1) 1 (2) 2
(3) 3 (4) 4

3. What is the HCF of $(x^2+bx-x-b)$ and $x^2+x(a-1)-a$?
(1) x+b(b) (2) x+a
(3) x+1(d) (4) x−1

4. If the HCF of x^2+x-12 and $2x^2-kx-9$ is x−k, then what is the value of k?
(1) −3 (2) 3
(3) −4 (4) 4

(3) $b + 2c = 2a + d$ (4) $b - 2c = 2a - d$

10. What are the values of c when the HCF of $x^3 + cx^2 - x + 2c$ and $x^2 + cx - 2$ over the rational is a linear polynomial?
(1) ± 1 (2) ± 2
(3) ± 3 (4) ± 4

11. If the HCF of $x^3 - 27$ and $x^3 + 4x^2 + 12x + k$ is a quadratic polynomial, then what is the value of k ?
(1) 27 (2) 9
(3) 3 (d) − 3

12. What is the LCM of $3(a^3 - b^3)$ and $11(a^4 - b^4)$?
(1) $33(a^3 - b^3)(a^2 + b^2)(a + b)$
(2) $(a^3 - b^3)(a^2 + b^2)(a - b)$
(3) $33(a - b)(a^2 + b^2 - ab)(a^3 - b^3)$
(4) $33(a^3 - b^3)(a^4 - b^4)$

13. I. The HCF of $x + y$ and $x^{10} - y^{10}$ is $x + y$.